PROPOSED ARCHITECTURE AND PROJECT PLAN

30-Day Hospital Readmission Predictive & Assistant System

1. Problem and Data

Task: Predict 30-day hospital readmission risk upon patient discharge and provide explanatory clinical insights using LLMs/SLMs + SHAP values.

Dataset:

- Structured EHR data (demographics, lab results, medications, previous admissions, discharge summaries).
- Unstructured medical notes and clinical guidelines for RAG contextualization.

Ingestion Pipeline:

- Ingest clinical text/notes through a pipeline using chunking and deduplication utilities (similar to labkit workflows).
- Continuously chunk and index new discharge summaries into a local vector storage (ChromaDB) for retrieval.

2. Golden Dataset

Creation Strategy:

- Define 30–50 representative ground-truth queries and clinical scenarios (e.g., "Why is patient X marked as high risk?", "What follow-up actions are recommended for a diabetic patient with elevated readmission risk?").
- Annotate queries with expected source document references (gold sources/citations) and ground-truth expert responses.

Evaluation Metrics:

- Retrieval Quality: Hit Rate @ 1, Hit Rate @ 3, and MRR on golden context passages.
- Answer/Generation Quality: Factuality, adherence to evidence (absence of hallucinations/unsupported claims), and proper citation of patient/guideline references.

3. RAG Design

- Chunking Strategy: Semantic chunking by section (e.g., discharge diagnosis, treatment plan, lab trends) with an approximate size of 90–120 words and 15–20 word overlap to preserve context across facts.
- Embedding Model: text-embedding-nomic-embed-text-v1.5 (or equivalent lightweight local embedding model) running via LM Studio to map text into vector space.
- Vector Database: ChromaDB (persistent local vector store with Cosine similarity space).

- RAG Pipeline Approach: Corrective / Agentic RAG approach. First retrieve top-k passages. A grader node assesses context sufficiency; if insufficient or partial, a query-rewriter generates an optimized search before generating the final response.

4. Agent and Tools (MCP Server)

- Agent Decision Logic: An SLM agent built with LangChain/LangGraph orchestrates actions based on user/clinician requests.
- Tools exposed via Model Context Protocol (MCP) Server:
    1. get_patient_readmission_risk(patient_id: str): Fetches ML model prediction (risk score + top 3 SHAP feature values).
    2. query_clinical_notes(patient_id: str, query: str): Vector search over patient history and discharge summaries.
    3. get_guidelines(topic: str): RAG lookup for clinical follow-up guidelines.
- Human-in-the-loop & Safety: Read-only lookups run directly; any high-impact actions (e.g., creating discharge follow-up tasks) require explicit clinician confirmation before execution.

5. Model Choices

- Predictive Model: XGBoost (with Logistic Regression baseline) for calculating the tabular readmission risk score and extracting SHAP feature importance values.
- Chat SLM: Local instruction-tuned Small Language Models (e.g., Qwen2.5-7B-Instruct-GGUF or Llama-3.2) hosted on LM Studio via local OpenAI-compatible API endpoints (localhost:1234).
- Rationale: Ensures patient data privacy (on-premise local inference), fast latency, zero external API costs, and structured tool calling support.

6. Timeline and Weekly Plan (Week 6 to Week 14)

- **Week 6: Architecture & Repo Setup**
    - Finalize docs/ architecture document, RFC review, environment setup.
- **Week 7: Data Preprocessing & Baseline**
    - Train baseline Logistic Regression & XGBoost model on EHR dataset; generate initial SHAP values.
- **Week 8: Golden Dataset & Ingestion**
    - Build golden dataset (gold_questions.jsonl); set up document ingestion and ChromaDB collection.
- **Week 9: RAG Pipeline Implementation**
    - Implement semantic search, test chunk sizes, evaluate Hit Rate @ 1 & @ 3 on golden dataset.

- **Week 10: Tool & MCP Server Integration**
    - Develop MCP server tools (get_patient_readmission_risk, RAG lookup); test tool validation schemas.
- **Week 11: LangGraph Agent Workflow**
    - Implement agent loop with LangGraph; integrate state validation, human-in-the-loop approval checks.
- **Week 12: Evaluation & Prompt Experiments**
    - Run systematic experiments across prompt techniques (Zero-shot, Few-shot, Corrective RAG); compute metric scores.
- **Week 13: Integration & UI / Polish**
    - End-to-end integration tests, error handling, refinement of clinical explanations.
- **Week 14: Final Presentation & Demo**
    - Final presentation, project report, repository documentation.

Diagram

